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Profile Summary

IEEE-published AI/ML Engineer specialising in carbon-efficient LLM inference — achieving 58% lower CO₂ emissions and 56% faster inference through self-pruning and activation freezing. Shipped 3+ production-grade AI systems spanning RAG pipelines, AI agents, and computer vision, including a voice interview platform that cut manual screening time by 70% and a mobile vision app with 92% detection accuracy. Skilled in PyTorch, LangChain, FAISS, fine-tuning (LoRA/QLoRA), and end-to-end model deployment from research to production.

Achievements

Best Final Year Project Award – Eco-Transformers

Achieved 58% lower CO₂ emissions and 56% faster LLM inference via self-pruning and activation freezing; published at ICNGCS 2025 (IEEE Xplore).

Publications

Eco-Transformers: Carbon-Efficient and Accelerated Inference through Self-Pruning and Activation Freezing in LLMs

Aug 2025

3rd International Conference on Next Generation Computing Systems (ICNGCS-2025), PSG iTech, Coimbatore.

DOI: [10.1109/ICNGCS64900.2025.11183217](https://doi.org/10.1109/ICNGCS64900.2025.11183217)

Experience

Innovation Engineer Trainee | *Forge Innovation & Ventures, Coimbatore*

Jan 2025 – Jul 2025

- Built an offline, voice-driven AI inventory assistant using Whisper STT and a quantized on-device LLM with FAISS-indexed search; reduced manual stock-lookup time by ~45% across 3 warehouse pilots with zero internet dependency.

Intern – Data Networking & Infrastructure | *Bharat Sanchar Nigam Limited (BSNL)*

Jun 2022 – Jul 2022

- Observed and analysed real-time signal routing and data transmission patterns across live optical fiber network nodes, building intuition for large-scale data pipeline behaviour and latency characteristics.

Projects

TestPilot AI – Autonomous AI Test Engineer Platform

[GitHub](#)

- Built an AI-powered software testing platform automating UI, performance, security, accessibility, and synthetic monitoring tests end-to-end; reduced total QA setup time by ~50% compared to manual configuration.
- Designed AI-driven QA workflows using Groq LLM-based agents for automated bug reporting, root cause analysis, and intelligent test recommendations — cutting manual triage effort by ~60% and detecting 3 critical bugs automatically across 200+ test cases in pilot testing.

Technologies Used: Python, PostgreSQL, Groq LLM, AI Agents, LangChain, Performance Testing, Stress Testing.

AI Book Scanner

[APK](#)

- Engineered a Flutter mobile app using a fine-tuned ConvNeXt vision model for book-cover recognition, achieving 92% detection accuracy across 1,000+ test images; deployed ML inference via FastAPI supporting 100+ concurrent users.

Technologies Used: Flutter, Firebase, FastAPI, PyTorch, ConvNeXt, FAISS, VLM, LLM, Transfer Learning.

AI Recruiter

[GitHub](#)

- Architected an end-to-end voice interview platform using Vapi and OpenRouter LLMs for real-time candidate Q&A, dynamic question generation, and automated evaluation scoring.
- Ran 50+ mock interview sessions during pilot testing, cutting manual HR screening time by 70% versus a fully manual process; integrated BlazeFace-based computer vision proctoring at 30 FPS (<50ms latency).

Technologies Used: Next.js, JavaScript, OpenRouter API, Supabase, BlazeFace, Vapi, Clerk, Computer Vision.

Technical Skills

Core Skills: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, RAG, AI Agents, Fine-tuning.

Programming: Python, SQL, TypeScript, C++, R

Frameworks & Libraries: PyTorch, Transformers, LangChain, OpenCV, FastAPI, Flask, TensorFlow, LoRA, QLoRA

Tools & Platforms: Firebase, Supabase, Google Cloud, n8n, HuggingFace Hub

Others: Flutter, Next.js, API Integration, Model Deployment, QA Automation

Education

B.Tech in Artificial Intelligence and Data Science (*Lateral Entry*)

2023 – 2026

Kumaraguru College of Technology, Coimbatore | CGPA: 8.02/10

Diploma in Computer Engineering

2020 – 2023

Tamilnadu Government Polytechnic College, Madurai | CGPA: 8.0/10